

ABHIJITH M. T.

Thermal Energy Storage & Solar Thermal Energy Researcher

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RESEARCH PROFILE

Thermal energy storage and solar thermal energy researcher with a strong foundation in physics, heat transfer, and thermal-fluid principles, combined with hands-on skills in experimental and computational laboratory settings. Completed Ph.D. in Physics – Green Energy Technology, specializing in solar-assisted latent heat thermal energy storage. My research interests lie in phase change material (PCM)-based thermal energy storage, high-temperature TES materials and heat exchangers, solar concentrating collectors and solar thermal devices, eutectic PCMs for cold storage and thermal comfort, and CFD-based numerical energy models.

- Currently serving as Project Associate at the Central Power Research Institute (CPRI), Prof. Sir C.V. Raman Road, Sadashiva Nagar, Bengaluru – 560080, Karnataka, India (Ministry of Power, Government of India), on the project “Development of High Temperature Thermal Energy Storage Materials and Compatible Heat Exchanger for CSP Application” (Ref. No. CPRI/R&D/TC/TH/2026).
- Experience in developing and characterizing PCM-based thermal energy storage systems with a focus on thermal performance and efficiency improvement.
- Professional in designing and executing experiments aimed at advancing solar thermal devices, heat exchangers, and thermal energy storage systems.
- Extensive experience in advanced experimental and analytical techniques, including ANSYS Fluent-based numerical and CFD simulation, for comprehensive thermal performance analysis.
- Skilled at communicating findings through detailed technical reports and peer-reviewed publications.
- Published 2 high impact research articles in international journals, with 2 further manuscripts under review and 1 book chapter.
- Committed to advancing sustainable energy solutions through high-performance thermal energy storage technologies.

EDUCATION

Degree	Specialization	Institution	Year	Result
Ph.D.	Physics (Green Energy Technology)	Pondicherry University	2026	Thesis submitted
M.Sc.	Physics	Maharaja's College, Ernakulam	2017	70%
B.Ed.	Physical Sciences	College of Teacher Education, M.G. University	2019	9.35 CGPA
B.Sc.	Physics	Nirmala College, Muvattupuzha, M.G. University	2015	86.5 CGPA

Ph.D. Thesis: “Development of Solar-Assisted Latent Heat Thermal Energy Storage System for Combined Heating and Cooling Applications,” Pondicherry University, 2026. Thesis submitted; institutional Thesis Submission Certificate available.

RESEARCH & PROFESSIONAL EXPERIENCE

- Project Associate** | Central Power Research Institute (CPRI), Bengaluru, India May 2026 – Present
- Project: “Development of High Temperature Thermal Energy Storage Materials and Compatible Heat Exchanger for CSP Application,” awarded by CPRI under the RSOP scheme (Ref. No. CPRI/R&D/TC/TH/2026).
 - Conducting research on high-temperature thermal energy storage materials and compatible heat exchanger development for concentrated solar power (CSP) applications.
- Ph.D. Researcher** | Department of Green Energy Technology, Pondicherry University 2021 – 2026
- Doctoral research on solar-assisted latent heat thermal energy storage for combined heating and cooling applications.
 - Conducted experimental, thermal-performance, and CFD investigations of PCM-based shell-and-tube heat exchangers, solar parabolic trough collectors, and eutectic PCMs.

PUBLICATIONS

Journal Articles

- [1] Abhijith, M. T., and Sreekumar, A., "Development of palmitic acid-lauryl alcohol as binary eutectic for cold thermal energy storage in buildings indoor thermal comfort application – Thermophysical studies and discharging characteristics," *Journal of Energy Storage*, 64, 107002, 2023. (IF: 10.7)
- [2] Vadakkan, S., Abhijith, M. T., and Sreekumar, A., "Thermophysical studies, corrosion analysis, and discharging characteristics of lauryl alcohol/lauric acid eutectic mixture as a potential phase change material for human comfort applications," *Energy Storage*, 5, e438, 2023. (IF: 4.0)

Manuscripts Under Review

- [1] Abhijith, M. T., Sreekumar, A., and Krishnamurthy, M., "Transient CFD Modelling and Experimental Investigation of Phase Change Heat Transfer in a Solar-Driven PCM-Based Shell-and-Tube Heat Exchanger Using a Representative Tube Model," *Energy Conversion and Management (Elsevier)*, Manuscript ID: ECM-D-26-06799. (IF: 10.9)
- [2] Abhijith, M. T., and Sreekumar, A., "Solar-Assisted Latent Heat Thermal Energy Storage System for Combined Heating and Cooling: Design and Performance Evaluation," *Solar Energy (Elsevier)*, Manuscript ID: SEJ-D-26-02453. (IF: 6.6)

Book Chapter

- [1] Abhijith, M. T., and Sreekumar, A., "Latent Heat Storage Materials and Applications," in *Thermal Energy – Applications, Innovations and Future Directions*, CRC Press, Taylor & Francis, 2023.

Conference Presentations & Workshops

- [1] Abhijith M. T. and Sreekumar A., "Investigation of a Solar-Assisted Latent Heat Thermal Energy Storage System Integrated with Shell and Tube Heat Exchanger – A Real-Time Performance Analysis and Discharge Characteristics," *ICOEST-2022*, Uttaranchal University, India and ENEA, Italy, Oct 2022.
- [2] Abhijith M. T. and Sreekumar A., "Development of Palmitic Acid–Lauryl Alcohol as Binary Eutectic for Cold Thermal Energy Storage in Buildings Indoor Thermal Comfort Applications," *National Conference on Recent Trends in Green Energy Technologies*, Pondicherry University, Dec 2022.
- [3] Abhijith M. T. and Sreekumar A., "Development of a Solar-Assisted Thermal Energy Storage System Integrated with Shell and Tube Heat Exchanger," *National Conference on Green Energy Technologies for Sustainability*, Mar 2024.
- [4] Abhijith M. T. and Sreekumar A., "Performance Evaluation of Nano-Engineered Phase Change Materials for Passive Thermal Management of Electronic Systems," *NCGETS-2025*, Mar 2025.
- [5] Abhijith M. T. and Sreekumar A., "Latent Heat Thermal Energy Storage for Sustainable Cooling and Shelf-Life Extension of Perishable Food Products," *International Conference on Energy and Water for Sustainable Environment*, Sep–Oct 2025.
- [6] Abhijith M. T., one-day expert workshop on "Role of Additive Manufacturing Applications Towards Sustainability," *Pondicherry University in collaboration with NCAM, Hyderabad*, Aug 2025.

RESEARCH INTERESTS

- Thermal Energy Storage and Phase Change Materials (PCMs)
- Solar Thermal Energy Systems and Solar Thermal Devices
- High-Temperature Thermal Energy Storage Materials and Heat Exchangers
- PCM-Based Shell-and-Tube Heat Exchangers and Heat-Transfer Enhancement
- Eutectic PCMs for Cold Thermal Energy Storage and Indoor Thermal Comfort
- Solar-Assisted Heating, Cooling, and Vapour Absorption Refrigeration
- Computational Heat-Transfer Modelling with ANSYS-Fluent

TECHNICAL SKILLS

- **Thermal Energy Storage:** PCM selection/characterization, melting-solidification, charging/discharging, hot/cold TES, eutectic PCM, thermal performance evaluation.
- **Solar Thermal Devices:** Parabolic trough collectors, solar-assisted thermal storage, heat-transfer fluids, collector performance analysis.
- **Numerical Simulation:** ANSYS Fluent; TES numerical simulations; transient analysis; PCM phase transition; solar thermal device modelling.
- **Experimental & Analysis:** Thermophysical characterization, thermal cycling, corrosion analysis, temperature/energy measurements, engineering calculations.

TEACHING EXPERIENCE

- Government Guest Lecturer – Physics, Sree Sankara Vidyapeetom College, Valayanchirangara, Kerala (Jun 2019 – Mar 2020).
- High School Physics Teacher, Government HSS Kallil, Methala, Kerala (Oct 2016 – Mar 2017).

ACHIEVEMENTS & CERTIFICATIONS

- Kerala Teachers Eligibility Test (KTET) – Category III (HS), Qualified.
- Certified on the Government Guest Lecturer panel for Physics by the Deputy Directorate of Collegiate Education, Government of Kerala (2019).
- National Level NCC Cadet with “C” Rank Certification, 18(K) BN NCC; recipient of the “Best Armed Performer” award.

REFERENCES

Dr. Prasanth Ravindran Professor & Head, Nanophotonics Lab Department of Green Energy Technology UNESCO Madanjeet School of Green Energy Technologies Pondicherry University, Puducherry – 605014, India Email: prasanth.ravindran@gmail.com Phone: +91-7639846141	Dr. Perumal Elumalai Professor & Lead, Electrochemical Energy Lab Department of Green Energy Technology UNESCO Madanjeet School of Green Energy Technologies Pondicherry University, Puducherry – 605014, India Email: drperumalelumalai@gmail.com Phone: +91-9751645895
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DECLARATION

I hereby declare that the above furnished information is true to the best of my knowledge.

Date: 11/08/2026

Place: Puducherry, India


ABHIJITH M. T.